Units & Dimensions

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Units

Science is about observations and related experiments. But we would appreciate scientific observations if it would provide us with more accurate and detailed information.

We will come across a lot of terms used to define physical quantities in Physics like length of an object, mass of an object, speed of a body, temperature of a body, magnetic field in a region, and many more. All these terms have a precise meaning and are used by scientists and engineers to communicate with each other.

When a physicist uses a term such as "kinetic energy," then it's meaning is understood by others immediately.

All the terms used in Physics represent a physical quantity which can be measured and compared to a standard reference.

For example, if a measurement of the length of a rod is taken as 10.5 meters, then it means that the length measured is 10.5 times the value accepted for a standard length defined to be "1 meter".

This chapter deals with the measurements of various physical quantities, their units and errors in their measurements. We will also discuss significant figures in measured values, and how dimensional analysis help us to understand the physical behavior of a quantity.

When we measure a quantity, we are finding how small or large the quantity is, when compared to a standard reference.

This standard reference is called the unit of the given physical quantity.

We can choose an arbitrary unit for a quantity, provided it is acceptable internationally.

To express any measurement, two things are needed:

(i) The unit for the quantity to be measured.

(ii) Magnitude of the quantity.

$\therefore$ Measurement of a quantity = Numerical Value × Unit

e.g., Length of a rod = 20 m

Mass of a box = 25 kg

The number of quantities observed in Physics are many, but all the quantities are directly or indirectly made from 7 fundamental quantities.

Physical quantities have two types:

(1) Fundamental or Base Quantities

Base or fundamental quantities have no dependency on other quantities.

These quantities are usually not defined in terms of other quantities.

The 7 fundamental quantities are shown in the table given below.

Fundamental Quantity	SI Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

(2) Derived Quantities

Derived quantities are those whose units can be written as a combination of fundamental quantities.

All other quantities different from the seven base quantities are derived quantities e.g., force, acceleration, kinetic energy, etc.

The units of derived quantities are called derived units.

For example, speed of an object is given by the relation

$$\text{Speed} = \frac{\text{Distance travelled}}{\text{Time taken}}$$

$$\therefore \quad \text{Unit of speed} = \frac{\text{Unit of distance}}{\text{Unit of time}} = \frac{\text{m}}{\text{s}}$$

Other units can also be derived in a similar manner.

Systems of units

Many unit systems are in use in different countries in world. Some widely used systems are shown in below table.

Unit system	Base unit for length	Base unit for mass	Base unit for time
MKS System	Meter	Kilogram	Second
CGS System	Centimeter	Gram	Second
FPS System	Foot	Pound	Second

In SI system, which is modern form of metric system, the 7 fundamental quantities are defined as:

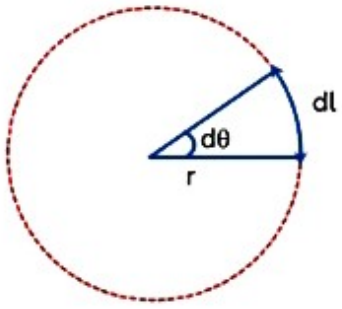
1. **Meter (m) :** 1 meter is defined as the path length which light travels in vacuum in a time interval of $\frac{1}{299,792,458}$ of a second. This definition was adopted in 1983.
2. **Kilogram (kg) :** The definition of kilogram was changed recently in 2019. Kilogram is defined in terms of Planck constant which is a fundamental constant in nature.
3. **Second (s) :** The time duration to complete 9,192,631,770 periods of the radiation of transition in between 2 hyperfine ground state levels of cesium-133 atom is known as 1 second.
4. **Ampere (A) :** The definition of 1 ampere is given by taking elemental charge value to be $1.602176634 \times 10^{-19}$ when expressed in unit of speed of light (c). This definition was also introduced in 2019.
5. **Kelvin (K) :** The new definition of Kelvin is defined by fixing the numerical value of the Boltzmann constant as 1.380649 × 10-23 J K-1.
6. **Mole (mol) :** One mole is defined as containing exactly $6.02214076 \times 10^{23}$ (the Avogadro constant) elementary entities.
7. **Candela (cd) :** The luminous intensity of a light source in a given direction emitting monochromatic radiation having a frequency of 540×10^{12} Hz and a radiant intensity of $\frac{1}{683}$ watt per steradian in that direction is defined as 1 candela.

There are two more supplementary SI units for plane angle and solid angle.

(i) Plane Angle : Plane angle is the ratio of the arc length dl of the circle to its radius r.

Plane angle, $d\theta = \frac{dl}{r}$ SI unit- Radian (rad)

A circle subtends a total plane angle of 2π radian at its centre.



Another unit for measurement of angle is degree.

$$1^\circ = \frac{\pi}{180}\text{rad}$$

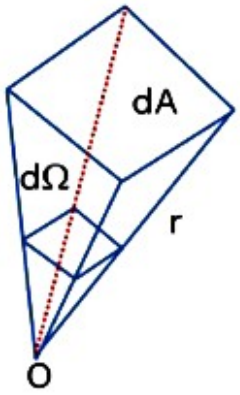
One degree is also equal to 60 minutes and 1 minute is equal to 60 seconds.

Points to remember

These minute and seconds are different than the ones used to represent time.

$$1^\circ = \frac{\pi}{180}\text{rad} = (60)' = (3600)''$$

(ii) Solid Angle : The angle which a spherical surface subtends at the centre of the sphere is called a solid angle.



Solid angle by a small area dA is given by,

$$d\Omega = \frac{dA}{r^2}$$

SI unit of a solid angle is steradian and is represented as 'sr'.

Supplementary units		
Plane Angle	radian	rad
Solid Angle	steradian	sr

The SI units for all other physical quantities can be derived from the SI base units mentioned above.

Some derived SI units are given special names.

e.g., SI unit of force is kg m s^{-2}. It is also called "newton".

There are also special names for some quantities like joule (J) for energy, watt (W) for power, coulomb (C) for charge, volt (V) for potential difference, tesla (T) for the magnetic field strength etc.

e.g., SI unit of work can also be written as "newton metre" or N m.

Similarly, SI unit of power can be written as "joule per second" or J s^{-1}.

A few more such units are : Vm^{-1} for electric field strength, Cm for dipole moment, N m^{-1} for surface tension etc.

Q. **Write the SI unit of density of a substance?**

Sol. Mass density $= \dfrac{\text{Mass}}{\text{Volume}}$

$\therefore$ SI unit of density $= \dfrac{\text{SI unit of mass}}{\text{SI unit of volume}} = \dfrac{\text{kg}}{\text{m}^3}$

Q. **Write the SI units of speed, acceleration and force.**

Sol. Speed, $v = \dfrac{\text{Distance}}{\text{Time}} = \dfrac{\text{m}}{\text{s}}$

Acceleration, $a = \dfrac{\text{Speed}}{\text{Time}} = \dfrac{(\text{m}/\text{s})}{\text{s}} = \dfrac{\text{m}}{\text{s}^2}$

Force, $F = \text{mass} \times \text{acceleration} = \text{kg} \times \dfrac{\text{m}}{\text{s}^2} = \text{kg m s}^{-2} = \text{N}$

Measurements

Measurement of time

For a long time, rotation of Earth about its axis was used to determine the length of day.

Repetitive systems like spring mass system, oscillating pendulums, quartz crystals are used to measure time.

For more accuracy and precision, atomic clocks are used.

Range of Time Intervals: The given table gives us an idea of how wide the range of time intervals for various events is.

Some Measured Time Intervals

Time interval	Seconds
Lifetime of proton	$> 10^{40}$
Half-life of double beta decay of ^{82}Se	3×10^{27}
Age of universe	5×10^{17}
Age of pyramid of Cheops	1×10^{11}
Time of Earth's orbit around the Sun (1 year)	3×10^{7}
Time of Earth's rotation about its axis (1 day)	9×10^{4}
Time between normal heartbeats	8×10^{-1}
Period of oscillation of 3-cm microwaves	1×10^{-10}
Typical period of rotation of a molecule	1×10^{-12}
Shortest light pulse produced (1990)	6×10^{-15}
Lifetime of least stable particles	$< 10^{-23}$
	*Approximate values

Measurement of length

To measure different range of lengths, different methods are used.

1. A metre scale can measure lengths between 10^{-3} m to 10^{2} m.
2. For smaller distances of range 10^{-4} m, we use Vernier calipers (explained in detail in Chapter "Experimental Physics").
3. For further small ranges like of the order of 10^{-5} m, we can use Screw gauge (explained in detail in Chapter "Experimental Physics").

   For very large and very small lengths as compared to above mentioned ranges, we use following methods.

Measurements of very small distances like Molecular Size:

Some instruments which can measure very small lengths are-

(a) Optical Microscope: It uses visible light, and it can resolve particles only up to the size of 10^{-7} m.

(b) Electron Microscope: Electron microscopes can resolve particles of size as small as 0.5×10^{-10} m by using electron beams focused by magnetic and electric fields.

(c) Tunnelling Microscopy: This method is used to estimate the size of molecules.

(d) Volumetric Method: Simple volumetric method can be used to find the size of large molecules e.g., oleic acid. Oleic acid is a soapy liquid.

Take 1 cm^3 volume of Oleic acid and mix it with 19 cm^3 alcohol. From this 20 cm^3 solution, take 1 cm^3 of the solution and again mix it with 19 cm^3 alcohol.

In this manner, we get an oleic acid concentration of $\frac{1}{400} cm^3$ in 1 cm^3 of alcohol.

Now some water is taken in a trough and some lycopodium powder is spread on it.

Now we drop 'n' drops (say) of oleic acid solution on the water surface. The drops spread on the water surface very fast. As the alcohol evaporates, a very thin, approximately circular film of oleic acid is left on the surface.

The area of the circular film can be measured by measuring its radius.

Volume of each drop can be calculated before dropping. Let it be V cm^3.

Amount of oleic acid in this solution $= \frac{nV}{20 \times 20} cm^3$

If t be the thickness of the film, then

$$A \times t = \frac{n \times V}{20 \times 20} cm^3$$

$$\boxed{t = \frac{n \times V}{A \times 400} cm}$$

The value of t is found to be of the order of 10^{-9} m.

We have used very dilute solution of oleic acid. So, thickness of the film is approximately equal to the molecular size of the oleic acid. Thus, the diameter of a molecule of oleic acid is equal to the thickness of the film and is about 10^{-9} m.

Q. **An oil drop having a radius of 0.25 mm spreads in the form of a circular film having a radius 5 cm on the water surface. Find the molecular size of the oil.**

Sol. Area of the film, $A = \pi r^2\ cm^2$

$= 25\ \pi\ cm^2$ [$\because$ Radius of the film, r = 5 cm]

Volume of the drop, $V = \frac{4}{3}\pi$ (Radius of oil drop)^3

$$= \frac{4}{3} \times \pi \times (0.25\,mm)^3$$

$$= \frac{4}{3} \times \pi \times (2.5 \times 10^{-1})^3\,mm^3$$

$$= \frac{4}{3} \times \pi \times 15.6 \times 10^{-3}\ mm^3$$

$$= \frac{4}{3} \times \pi \times 15.6 \times 10^{-6}\ cm^3$$

Molecular size = Thickness of the film

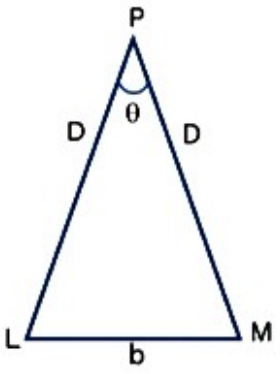
$$= \frac{V}{A} = \frac{4\pi \times 15.6 \times 10^{-6}}{3 \times 25\pi} = 0.83 \times 10^{-6}\ cm = 8.3 \times 10^{-9}\ m.$$

Measurement of Large interplanetary Distances

We use parallax method to find very large distances.

Parallax Method:

With a fixed reference in background, the apparent shift of a body's position when eyes are alternately opened and closed, is called parallax.



To explain further, consider a metal rod held in front of our eyes at some distance. Let some reference point, like a pole or a box, be there in the background.

On observing the rod, first with our right eye closed and then with our left eye closed, we can find a difference in the position of the rod with respect to the background in the two cases. This is called parallax.

In order to find the distance of a planet very far away by parallax method, the planet is observed from two different positions on earth using a telescope.

Consider the distance between two different positions on earth be "b".

The distance "b" is known as **basis**.

For a very far off planet,

i.e. LP = MP = D.

Also $b << D$

Thus, we can say that the length LM is approximately a circular arc with arc length b having its centre at P. The radius of the circle is D.

So, the angle θ can be written as,

$$\theta = \frac{b}{D}$$

$$\Rightarrow \boxed{D = \frac{b}{\theta}}$$

where θ = parallax angle or parallactic angle.
θ is taken in radian not in degree.

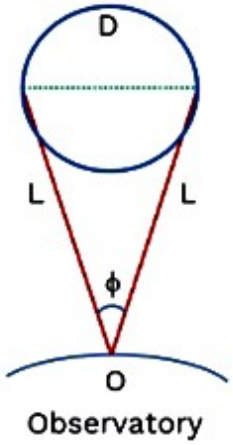
To find a planet size:

After knowing the distance of a planet, its diameter can also be measured.

To measure the diameter of a planet, its two diametric opposite ends are observed from a single point on earth.
If ϕ is the angle between the two lines L shown, then diameter of the planet can be written as,

$$D = L\phi$$

where ϕ = angular size or angular diameter of the planet.



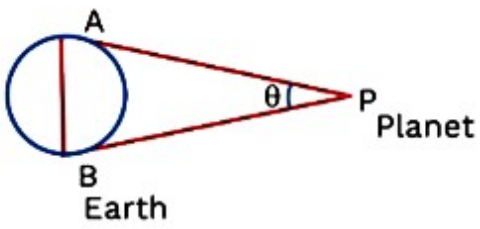
Q. **For a far off planet, the parallax angle from two diametric extremes on the earth's equator is 2.0 minutes.**
For earth's radius = 6400 km, what is the distance of the planet from earth?

Sol. Angle of parallax, θ = 2 minute

$$= 2 \times \left(\frac{1}{60}\right)^{\circ}$$

$$\Rightarrow \quad \theta = 2 \times \frac{1}{60} \times \frac{\pi}{180} \text{rad} \quad \left[\because 1^{\circ} = \frac{\pi}{180} \text{rad}\right]$$

$$\theta = \frac{\pi}{5400} \text{rad}$$



By parallax method,

$$BP = \frac{AB}{\theta}$$

$$\Rightarrow \quad BP = \frac{\text{Diameter of earth}}{\theta}$$

$$\Rightarrow \quad BP = \frac{2 \times 6400}{\left(\frac{\pi}{5400}\right)} \text{km}$$

$$\Rightarrow \quad BP = 2.2 \times 10^{7} \text{ km}$$

$$= 2.2 \times 10^{10} \text{ m}$$

Range of Lengths : The following table gives an idea of how wide the range of length is for the objects present in this universe.

Some Measured Lengths

LENGTH	IN METERS
Distance to the farthest observed quasar	2×10^{26}
Distance to the Andromeda galaxy	2×10^{22}
Radius of our galaxy	6×10^{19}
Distance to the nearest star (Proxima Centauri)	4×10^{16}
Mean orbit radius for the most distance planet (pluto)	6×10^{12}
Radius of the sun	7×10^{8}
Radius of the Earth	6×10^{6}
Height of Mt. Everest	9×10^{3}
Height of a typical person	2×10^{0}
Size of a typical virus	1×10^{-6}
Radius of a hydrogen atom	5×10^{-11}
Effective radius of a proton	1×10^{-15}
	*Approximate values.

There are some frequently used units for length also described in the table below.

PHYSICAL QUANTITY	UNIT NAME	SYMBOL	VALUE IN UNIT
Length	Fermi	F	10^{-15} m
	Angstrom	Å	10^{-10} m
	Astronomical Unit	AU	1.496×10^{11} m (Average distance of the Sun from the Earth)
	Light Year	ly	9.46×10^{15} m (Distance travelled by light in vacuum in a year)
	Parsec	Parsec	3.08×10^{16} m (3.26 ly)

Q. **Light year is unit of distance which is equal to the distance travelled by light in 1 year. Convert 1 light year into meters. Also, using the data from above table, find the distance of Proxima Centauri star in light years.**

Sol. $1 \text{ year} = 1 \text{ year} \times \frac{365 \text{ days}}{1 \text{ year}} \times \frac{24 \text{ hr}}{1 \text{ day}} \times \frac{60 \text{ min}}{1 \text{ hr}} \times \frac{60 \text{ s}}{1 \text{ min}} = 3.16 \times 10^{7} \text{ s}$

Speed of light, $c = 3.00 \times 10^{8}$ m/s.

Distance covered by light in 1 year, $D = (3.00 \times 10^{8} \text{ m/s})(3.16 \times 10^{7} \text{s}) = 9.48 \times 10^{15} \text{m}$,

So, 1 light-year $= 9.48 \times 10^{15}$m.

The distance to Proxima Centauri is

$$(4.0 \times 10^{16} \text{m}) \times \frac{1 \text{light} - \text{year}}{9.48 \times 10^{15} \text{m}} = 4.2 \text{light} - \text{years}$$

So, the light from the star Proxima Centauri reaches earth in 4.2 years.

Measurement of mass

Quantity of matter in a body is known as mass. Mass does not change with temperature, pressure, location etc.

A few tools to determine masses are given below.

(a) Mass Spectrograph for measuring small masses:

In upcoming chapters in class 12th, we will see that charged particles execute circular motion when moving in uniform magnetic fields and accelerate when kept in uniform electric fields. The radii of the circular paths are proportional to the mass of the particles. So, the value of the radius gives an estimate of the mass of the particles. This principle is used in mass spectrographs.

Though the SI unit of mass is kilogram (kg), using this unit is inconvenient while measuring molecular and atomic masses. Here we use another standard unit of mass called **atomic mass unit (u)**. It is defined as follows.

The atomic mass unit is the one twelfth of the mass of an atom of carbon-12 isotope $\left({}^{12}_{6}C\right)$ including the mass of electrons.

Thus, $1u = 1.66 \times 10^{-27}kg$

(b) Masses of stars and planets can be determined using Newton's law of gravitation. Day to day life masses can be measured using gravity balances, spring balances, digital balances etc.

Range of Masses : The following table gives us an idea to how wide the range of masses is for the objects present in this universe.

Some Measured Masses

OBJECT	MASS IN KILOGRAMS
Known universe (estimate)	10^{53}
Sun	2×10^{30}
Earth	6×10^{24}
Moon	7×10^{22}
Elephant	4×10^{3}
Person	6×10^{1}
Grape	3×10^{-3}
Speck of dust	7×10^{-10}
Virus	1×10^{-15}
Penicillin molecule	5×10^{-17}
Uranium atom	4×10^{-26}
Proton	2×10^{-27}
Electron	9×10^{-31}
	*Approximate values.

Dimensional Analysis

Dimensions of physical quantities

Every derived physical quantity can be expressed in form of the 7 fundamental quantities.

Fundamental quantities are represented using square brackets.

The 7 dimensions of the fundamental quantities are given in the below table.

QUANTITY	DIMENSION
Mass	[M]
Length	[L]
Time	[T]
Electric current	[A]
Temperature	[K]
Amount of substance	[mol]
Luminuous intensity	[Cd]

The dependence of all other physical quantities on these base quantities can be expressed in terms of their dimensions.

For example, the speed v, of a moving object is given by the relation $v = \frac{\text{Distance}}{\text{Time}}$

Thus, the dimension of speed can be given as

$$[v] = \frac{[L]}{[T]} = [LT^{-1}]$$

And, the dimension of acceleration can be given as

$$[a] = \frac{[L]}{[T^2]} = [LT^{-2}]$$

The dimensions $[LT^{-1}]$ just tells how the speed changes with change in length and time, it does not tell its magnitude or its unit.

It can be seen that speed of a body has one dimension in length and -1 dimension in time. As speed is independent of mass, so it has zero dimension in mass.

Thus, speed has the dimensions $[M^0LT^{-1}]$.

Dimensional equations and formulae

The expression of a physical quantity in terms of its dimensions is called its dimensional formula.

e.g. The dimensional formula of force is $[MLT^{-2}]$ and that for acceleration is $[M^0LT^{-2}]$.

Dimensional formulae for some more quantities are given in the table in end of the chapter.

An equation with the physical quantity on LHS and its dimensional formula on RHS, is called the dimensional equation of that quantity.

For example,

Speed, $[v] = [M^0 L T^{-1}]$

Area, $[A] = [M^0 L^2 T^0]$

Force, $[F] = [M L T^{-2}]$

Principle of dimensional homogeneity

The principle of dimensional homogeneity states that we cannot add or subtract two quantities with different dimensional formulae.

So, speed and force cannot be added or mass and density cannot be subtracted.

Thus, it is to be noted that if A + B and A – B are meaningful, then A and B have same dimensions and same nature.

If A = B is correct, then A and B have same dimensions and same nature.

This concept is applicable for checking the correctness of a mathematical equation.

Whenever there is a multiplication or division of 2 quantities, then the dimensions of the resulting quantity is calculated by multiplying or dividing the dimensions of the quantities undergoing the multiplication or division.

For example, Force = Mass × Acceleration

Thus, the dimensions of force ⇒ [Force] = [Mass] × [Acceleration]

$= [M] [LT^{-2}] = [M L T^{-2}]$

Applications of dimensional analysis

a. To check whether an equation is correct dimensionally

For a mathematical equation to be correct, the dimensional formula of all the terms in it should be same. This is known as "dimensional homogeneity principle".

Dimensional homogeneity principle is used to check whether a mathematical relation between quantities is correct or not.

Every correct mathematical formula always satisfies principle of dimensional homogeneity. But its inverse is not true.

It means if an equation is correct dimensionally correct, then it does not guarantee that the equation is physicaly correct as well.

For example, consider the following equation of area of a circle with radius r,

$$\boxed{A = \frac{\pi}{12} r^2}$$

Checking the dimensions on both sides, we get

$[A] = [M^0 L^2 T^0]$ ...(L.H.S.)

$$\left[\frac{\pi}{12} r^2\right] = \left[M^0 L^2 T^0\right]$$

$= [M^0 L^2 T^0]$ ...(R.H.S)

We can see that the dimensions are same for LHS and RHS, but we know that area of a circle is πr^2.

Thus, we can say that if an equation fails the dimensional homogeneity principle, then it is wrong, but even if it passes, it is not proved to be right.

This is the limitation of dimensional homogeneity principle. It cannot test for the constant multiples in an equation.

Q. **From the equations given below, check which are correct dimensionally as well as physically. v is final velocity, u is initial velocity, a is acceleration and s is displacement.**

(i) $v_{avg} = \frac{u+v}{3}$

(ii) $s = ut + \frac{1}{2}at^2$

(iii) $v^2 - u^2 = \frac{2s}{a}$

Sol. (i) Checking the dimensions on both sides,

Average velocity, $[v_{avg}] = [LT^{-1}] \rightarrow$ L.H.S.

Initial velocity, $[u] = [LT^{-1}]$

Final velocity, $[v] = [LT^{-1}]$ $\rightarrow$ R.H.S.

All terms in LHS and RHS have same dimensions. So, the equation is correct dimensionally.

But it is not correct physically as average velocity for given case will be,

$$v_{av} = \frac{u+v}{2}$$

(ii) $$s = ut + \frac{1}{2}at^2$$

$[s] = [L] \rightarrow$ L.H.S.

$[u\,t] = [u]\,[t]$

$= [LT^{-1}]\,[T]$

$[u\,t] = [L] \rightarrow$ R.H.S.

$$\left[\frac{1}{2}at^2\right] = \left[LT^{-2}\right]\left[T^2\right] = \left[L\right] \rightarrow \text{R.H.S.}$$

All the terms have same dimensional formula. So, the equation is correct dimensionally. It is also physically correct as per the equation of motion for constant acceleration.

(iii) $v^2 - u^2 = \frac{2s}{a}$

Checking the dimensions on both sides.

$[v^2] = [LT^{-1}]\,[LT^{-1}]$

$= [L^2T^{-2}]$ → L.H.S.

$[u^2] = [LT^{-1}]\,[LT^{-1}]$

$= [L^2T^{-2}]$

$$\left[\frac{2s}{a}\right] = \frac{[L]}{[LT^{-2}]} = [T^2] \rightarrow \text{R.H.S.}$$

All terms have not the same dimensional formulae. So, this equation is incorrect.

Q. **The relation between the mass m_1 kept on a surface of area A and the pressure P on the given area is given by**

$$P = \frac{(m_1 + m_2)x}{A}$$

What are the dimensions of x and m_2?

Sol. Dimensions of LHS and RHS should be same.

So, $$[P] = \left[\frac{(m_1 + m_2)\,x}{A}\right] \quad \ldots(i)$$

Here, m_1 and m_2 are added. Addition can be performed on quantities of same dimensions. So, m_2 is having dimension of mass.

$$[m_2] = [m_1] = [M]$$

From equation (i), we have

$$[P] = \frac{[m_1 + m_2][x]}{[A]}$$

$$\Rightarrow \quad [ML^{-1}T^{-2}] = \frac{[M][x]}{[L^2]}$$

$$\Rightarrow \quad [ML^{-1}T^{-2}] = [ML^{-2}][x]$$

$$\Rightarrow \quad \frac{\left[ML^{-1}T^{-2}\right]}{\left[ML^{-2}\right]} = \left[x\right]$$

$$\Rightarrow \quad \left[x\right] = \left[LT^{-2}\right]$$

Thus, the quantity x is acceleration.

Q. **The distance covered by a particle in time t is given by**
$x = a + bt + ct^2 + dt^3$
Find the dimensions of a, b, c and d.

Sol. Since the given relation must be dimensionally correct, so all the terms appearing in the equation must have same dimensions.
So, dimension of a, [a] = Dimension of distance x = [L].
Dimension of bt, [b T] =[L],
Then, [b] = [L T^{-1}].
Dimension of ct^2, [c T^2] = [L]
Then, [c] = [L T^{-2}].
Dimension of dt^3, [d T^3] = [L]
Then, [d] = [L T^{-3}].

Q. **Check the dimensional correctness of the formula:**

$$s_n = u + \frac{a}{2}\left[n - \frac{1}{2}\right]$$

Here S_n is the displacement during n^{th} second, u is initial velocity, a is acceleration.

Sol. If we go by our standard method of finding the dimensions of the quantities in LHS and RHS, we will find that this formula is dimensionally incorrect as the dimension of u is [LT^{-1}] and dimension of s_n is [L] only.
However, the constant values appearing in this formula are values of time.
So, we have to see the dimensions of the constants in the derivation of this formula as well.
From 3rd equation of motion for constant acceleration,

We know that displacement after n seconds, $s = un + \frac{1}{2}an^2$

And, displacement after (n-1) seconds will be, $s' = u(n-1) + \frac{1}{2}a(n-1)^2$

So, the displacement during n^{th} second,

$$s_n = s - s' = \left(un + \frac{1}{2}an^2\right) - \left(u(n-1) + \frac{1}{2}a(n-1)^2\right)$$

$$s_n = \left(u\left[n - (n-1)\right] + \frac{1}{2}a\left[n^2 - (n-1)^2\right]\right)$$

$$s_n = \left(u[1] + \frac{1}{2}a\left[n^2 - (n-1)^2\right]\right)$$

The 1 multiplied to u is not a constant, infact it is 1 second so it has a dimension of time.

Also, on simplifying the term $\left[n^2 - (n-1)^2\right]$, we get $\left[n - \frac{1}{2}\right]$ which suggests that it has dimension of time [T] but it has dimension of [T^2] actually from the expression $\left[n^2 - (n-1)^2\right]$.

So, this equation is indeed dimensionally correct.
Here, the constants appearing in the equation were not dimensionless.

b. Finding relation between various physical quantities

We can use dimensional analysis to find the dependency of various physical quantities on each other.

Consider the given example.

We know the kinetic energy of a body depends upon

(i) Mass of the body
(ii) Velocity of the body

Let us consider that kinetic energy K depends on mass m and velocity v as

$K = c\, m^p v^q$ ...(i)

Here c is constant, having no dimensions. p and q are powers on m and v respectively.

If the above relation is correct, then terms on both sides should have the same dimensions.

i.e., $[K] = [c\, m^p v^q]$

$\Rightarrow \quad [K] = [c]\,[m^p]\,[v^q]$

$\Rightarrow \quad [M^1 L^2 T^{-2}] = [M^0 L^0 T^0]\,[M]^p\,[M^0 L^1 T^{-1}]^q$

$[M^1 L^2 T^{-2}] = [M^p L^q T^{-q}]$

On comparing the powers,

$p = 1$
and $q = 2$
Putting these values in equation (i), we have

$$K = c m v^2$$

We have to find the value of constant c by different methods.

In this case, $c = \frac{1}{2}$.

Hence the relation becomes $K = \frac{1}{2}mv^2$

Q. **When a solid sphere moves through a liquid, the liquid opposes the motion with a force F. The magnitude of F depends on the coefficient of viscosity η of the liquid, the radius r of the sphere and the speed v of the sphere. Assuming that F is proportional to different powers of these quantities, guess a formula for F using the method of dimensions. Given that $\eta = [M^1L^{-1}T^{-1}]$.**

Sol. Given that the opposing force (F) depends on η, r and v.
Let $F = k\eta^a r^b v^c$, where K is a dimensionless constant.

$$[F] = [\eta^a r^b v^c]$$

Writing dimensions on both sides,

$$[MLT^{-2}] = [[ML^{-1}T^{-1}]^a [L]^b [LT^{-1}]^c]$$

$$[MLT^{-2}] = [M^a L^{-a+b+c} T^{-a-c}]$$

Comparing the powers,

$a = 1$

$-a + b + c = 1 \quad \Rightarrow \quad -1 + b + c = 1$

$b + c = 2$

$c = 1$

So, $b + 1 = 2 \quad \Rightarrow \quad b = 1$

$\therefore \quad F = K \eta^1 r^1 v^1$

$$\boxed{F = K\eta r v}$$

The value of K is found to be 6π which we shall see in chapter "Mechanical properties of fluids".

Q. **For an object moving on a circular path, the centripetal force acting on the object towards the centre of the circle is found to be depending on the speed v of particle, mass m of particle and the radius r of the circular path. Derive an expression for the force F using dimensional analysis.**

Sol. Consider the force F to be,

$$F = k\, m^x v^y r^z .$$

where k = dimensionless constant.

So, we can write,

$$[F] = [m]^x [v]^y [r]^z$$

Writing the dimensional formulae on both sides,

$$\Rightarrow \quad [M^1L^1T^{-2}] = [M]^x [LT^{-1}]^y [L]^z$$

$$[M^1L^1T^{-2}] = [M^x L^{y+z} T^{-y}]$$

Comparing the powers,

$$x = 1$$

$$y + z = 1,$$

and, $y = 2.$

Then, $z = -1.$

So, we can write,

$$F = k\, mv^2r^{-1}$$

$$\text{or } F = \frac{k m v^2}{r}$$

In study of circular motion, we will see that k = 1.

$$\therefore \quad \boxed{F = \frac{mv^2}{r}}$$. This central force F is called centripetal force.

Special cases of dimensions

The quantities which don't have any dimensions are called dimensionless quantities. The relation among such quantities cannot be deduced using dimensional method. The quantities belonging to this category are:

(i) Pure numbers e.g. 1, 2, 3, 4, 5, 6 etc.

(ii) Arguments of logarithmic, exponential and trigonometric functions are dimensionless.

For example, let the position of a particle undergoing a periodic motion is given by

the relation, $x = A\sin(\omega t)$, then (ωt) whose sine function has been taken, should necessarily be a dimensionless quantity

$\Rightarrow$ $[\omega t] = [M^0L^0T^0]$. Thus if ω represents angular frequency, t should represent time

so that

$$[\omega]\,[t] = [M^0L^0T^0]$$

$$[T^{-1}]\,[T] = [M^0L^0T^0]$$

Same is true for the arguments of logarithmic and exponential functions.

(iii) Ratio of physical quantities having the same dimensions are dimensionless.

For example, plane angle (the ratio of two lengths), solid angle (area/r^2), refractive index, relative density, magnifying power (the ratio of image height and object height) etc. are dimensionless quantities.

Percentages and relative ratios are also dimensionless.

Dimensions for derivatives

The first derivative of a quantity with respect to another quantity has the same dimensions as that of the ratio of the first quantity to the second quantity.

$$\left[\frac{du}{dx}\right] = \left[\frac{u}{x}\right]$$

However, this scenario changes for higher order derivatives.

For 2nd order derivatives, the dimensions are given as,

$$\left[\frac{d^2u}{dx^2}\right] = \left[\frac{u}{x^2}\right]$$

In general, for n^{th} order derivative, $\left[\frac{d^nu}{dx^n}\right] = \left[\frac{u}{x^n}\right]$.

Dimensions for integrals

For an integral, the dimension of integral of one quantity wrt another quantity are same as the dimensions of the product of the two quantities.

$$\left[\int u \cdot dx\right] = [ux]$$

Q. Find dimensional formula for 4th derivative of pressure wrt position x.

Sol.

$$\left[\frac{d^4p}{dx^4}\right] = \left[\frac{p}{x^4}\right] = \left[\frac{F}{Ax^4}\right]$$

$$= \left[\frac{MLT^{-2}}{L^2 \times L^4}\right] = \left[ML^{-5}T^{-2}\right]$$

Q. Find the dimensional formula for

$$\psi = \int_2^5 E.dV$$

Where E is energy, V is volume.

Sol.

$$\left[\int E \cdot dV\right] = [EV] = \left[ML^2T^{-2} \cdot L^3\right]$$

$$= \left[ML^5T^{-2}\right]$$

Q. The interaction force between 2 atoms is expressed as the relation $F = \alpha\beta e^{-\left(\frac{x^2}{\alpha kT}\right)}$, where k= Boltzmann constant, T= Temperature and x is the distance between the atoms.
If α and β are constants, then find the dimensions of β. (Boltzmann constant = JK^{-1})

Sol. The given expression is,

$$F = \alpha\beta e^{-\left(\frac{x^2}{\alpha kT}\right)}$$

The term in the power of exponent should be dimensionless.

$$\frac{x^2}{\alpha kT} = [M^0L^0T^0]$$

$$\Rightarrow \frac{[L^2]}{[\alpha][ML^2T^{-2}K^{-1}][K]} = [M^0L^0T^0] \Rightarrow [\alpha] = [M^{-1}T^2]$$

Now, $[\alpha][\beta] = [F]$

$\Rightarrow [M^{-1}T^2][\beta] = [MLT^{-2}] \Rightarrow [\beta] = [M^2LT^{-4}]$

Q. The dimensional formula of coefficient of viscosity can be expressed as $M^aL^bT^c$. Find a, b and c.

Sol. Viscous force, $F = 6\pi\eta rv$

$$[\eta] = \left[\frac{F}{6\pi rv}\right] = \left[\frac{MLT^{-2}}{L \times LT^{-1}}\right] = \left[ML^{-1}T^{-1}\right]$$

Q. Which of the following quantities, ratios are dimensionless?

(i) $\frac{\text{Work}}{\text{Energy}}$

(ii) **sin(i)**

(iii) $\frac{\text{Momentum}}{\text{Time}}$

Sol. (i) Dimensions of work and energy are same i.e., $[ML^2T^{-2}]$. So, their ratio is a dimensionless quantity.

(ii) sin(i), Here i represents an angle.

An angle is dimensionless, so sin(i) is also dimensionless.

(iii) $\left[\frac{\text{Momentum}}{\text{Time}}\right] = \frac{[MLT^{-1}]}{[T]} = [MLT^{-2}]$

So, it is not dimensionless.

c. Changing the fundamental quantities

Q. If the speed of light (c), acceleration due to gravity (g) and pressure (P) are taken as fundamental units, then dimensions of gravitational constant (G) are

Sol. Here we have to find G in terms of c, g and P.

Let $G = c^x g^y P^z$

By Newton's universal gravitational law, $F = \frac{G m_1 m_2}{r^2}$

$$G = \frac{Fr^2}{m_1 m_2} = \frac{[MLT^{-2}][L^2]}{[M^2]} = [M^{-1}L^3T^{-2}]$$

Now, writing dimensions $G = c^x g^y p^z$

$$[M^{-1}L^3T^{-2}] = [LT^{-1}]^x [LT^{-2}]^y \left[\frac{F}{L^2}\right]^z$$

$$[M^{-1}L^3T^{-2}] = [LT^{-1}]^x [LT^{-2}]^y [ML^{-1}T^{-2}]^z$$

$$[M^{-1}L^3T^{-2}] = [M^z L^{x+y-z} T^{-x-2y-2z}]$$

Comparing the powers,

$z = -1$

$x + y - z = 3 \Rightarrow x + y - (-1) = 3 \Rightarrow x + y = 2$

$-x - 2y - 2z = -2 \Rightarrow -x - 2y - 2(-1) = -2 \Rightarrow x + 2y = 4$

$y = 2 - x$

$\Rightarrow x + 2(2 - x) = 4$

$x + 4 - 2x = 4$

$-x = 0$

$\boxed{x = 0}$

$\boxed{y = 2}$

So, $\boxed{G = c^0 g^2 P^{-1}}$

Q. **If area (A), velocity (v) and density (ρ) are base units, then the dimensional formula of force can be represented as-**

Sol. Let $F = A^x v^y \rho^z$

$$\left[MLT^{-2}\right] = \left[L^2\right]^x \left[LT^{-1}\right]^y \left[ML^{-3}\right]^z$$

$$\left[MLT^{-2}\right] = \left[M^z L^{2x+y-3z} T^{-y}\right]$$

Comparing powers,

$z = 1$

$-y = -2 \quad \Rightarrow \quad y = 2$

$2x + y - 3z = 1 \quad \Rightarrow \quad 2x + 2 - 3 = 1$

$2x = 2$

$\boxed{x = 1}$

So, $\boxed{F = A^1 v^2 \rho^1}$

Q. **Let us consider a system of units in which mass and angular momentum are dimensionless. If length has dimension of L, which of the following statement(s) is/are correct?**

(A) **The dimension of force is L^{-3}**

(B) **The dimension of power L^{-5}**

(C) **The dimension of energy is L^{-2}**

(D) **The dimension of linear momentum is L^{-1}**

Sol. **(A, C, D)**

Given that mass and angular momentum are dimensionless.

So, dimension of mass = dimension of angular momentum

or $M^1L^0T^0 = [M^1L^2T^{-1}]$

or $[L^2] = [T]$

Now, dimension of power, $[P] = [M\ L^2\ T^{-3}] = [\ L^2\ T^{-3}] = [L^2\ L^{-6}] = [L^{-4}]$

Dimension of energy, $[E] = [M\ L^2\ T^{-2}] = [\ L^2\ T^{-2}] = [L^2\ L^{-4}] = [L^{-2}]$

Dimension of force, $[F] = [M\ L\ T^{-2}] = [\ L\ T^{-2}] = [L\ L^{-4}] = [L^{-3}]$

Dimension of linear momentum, $[p] = [M\ L\ T^{-1}] = [\ L\ T^{-1}] = [L\ L^{-2}] = [L^{-1}]$

Miscellaneous Examples

Q. **In a new unit system, 1 unit of time is equal to 10 second, 1 unit of mass is 5 kg and 1 unit of length is 20 m. In the new system of units, 1 unit of energy is equal to-**

(A) 20 J **(B)** $\frac{1}{20}$ **J** **(B) 4 J** **(D) 16 J**

Sol. Energy, $E = ML^2T^{-2}$

$$1J = \frac{1kg(1m)^2}{(1s)^2} = \frac{kg\,m^2}{s^2}$$

In new units,

Energy, $$E' = \frac{(5kg)(20\,m)^2}{(10\,s)^2} = \frac{5\times 400}{100}\frac{kg\,m^2}{s^2}$$

$$E' = \frac{20\,kg\,m^2}{s^2}$$

$$E' = 20E$$

$$\boxed{E' = 20\,J}$$

Q. **In two systems of units, the relation between velocity, acceleration and force is given by,**

$$v_2 = \frac{v_1\,\epsilon^2}{\tau}, \quad a_2 = a_1\,\epsilon\,\tau, \quad F_2 = \frac{F_1}{\epsilon\,\tau}.$$

Where ϵ and τ are constants. Then in new system, find

(A) $\frac{m_2}{m_1}$ **(B)** $\frac{l_2}{l_1}$

Sol. (a) By Newton's 2nd Law, F = ma

so, $m_2 = \frac{F_2}{a_2}$ and $m_1 = \frac{F_1}{a_1}$

so, $$\frac{m_2}{m_1} = \left(\frac{F_2}{a_2}\right)\left(\frac{a_1}{F_1}\right) = \left(\frac{F_2}{F_1}\right)\left(\frac{a_1}{a_2}\right) = \left(\frac{1}{\epsilon\,\tau}\right)\left(\frac{1}{\epsilon\,\tau}\right) = \frac{1}{\epsilon^2\,\tau^2}$$

(b) Also, $L = \frac{v^2}{a}$

so, $L_2 = \frac{v_2^2}{a_2}$ and $L_1 = \frac{v_1^2}{a_1}$

$$\frac{L_2}{L_1} = \left(\frac{v_2}{v_1}\right)^2\left(\frac{a_1}{a_2}\right) = \left(\frac{\epsilon^2}{\tau}\right)^2\left(\frac{1}{\epsilon\,\tau}\right) = \frac{\epsilon^4}{\tau^2\cdot\epsilon\,\tau} = \frac{\epsilon^3}{\tau^3}$$

Error analysis

Any measuring device gives a reading with a certain uncertainty. This uncertainty is known as error in a measurement.

There can be an error in a measurement due to faulty measurement device or wrong skills of the person taking the measurement.

e.g., If thickness of a plate is being measured with help of a simple scale and it is 2.5 cm, then we can say that the thickness is between 2.4 cm and 2.6 cm. So, this measurement is uncertain by ± 0.1 cm.

Uncertainty is because a metre scale can only measure a length upto 0.1 cm. So, we can say the limit of resolution (or least count) for a metre scale is 0.1 cm.

There are several other factors which give rise to errors.

Few terms related to measurement are described below:

Accuracy: Closeness of the measured value and true value is called accuracy of measurement.

If the true value of a quantity is 2.85 and the measurement taken by a person A is 2.76 and another person B measures it to be 2.87, then measurement of person B is more accurate as the true value is close to 2.87.

Precision :

The limit or resolution to which measurement is being taken describes the precision of the measurement.

If a certain length is measured by a meter scale (least count 0.1 cm) and the same length is measured by vernier calliper (least count 0.01 cm), then the measurement by the Vernier calliper is more precise.

Greater precision does not always show greater accuracy.

Let a person measures a length of 15.75 cm by the above mentioned two devices, and obtains the values 15.7 cm and 15.45 cm respectively.

Although the first value is less precise, but it is more accurate as it is closer to the true value.

Errors are of two types:

1. **Systematic Errors**

   Errors arising in one direction only, i.e. either positive or negative direction are systematic errors.

   Error is positive when measured value is greater than true value.

   Error is negative when measured value is less than true value.

   Systematic errors can arise due to:

   (1) Instrumental Errors
      (i) Improper Design or Calibration
      (ii) Zero Error
   (2) Imperfect procedure for the experiment.
   (3) Personal Errors

2. Random Errors

These errors arise due to the reasons which are unpredictable during an experiment. e.g. random changes in pressure, temperature, voltage supply etc.

In order to reduce the errors in a measurement, a large number of observations are taken which reduces the mean value of the error after taking the arithmetic mean.

Least Count Error

The minimum value that can be measured by an instrument is known as its least count or resolution of the instrument.

We can reduce least count error in a measurement by using high precision measuring instrument and improved experimental techniques.

Absolute, Relative and Percentage Errors

Though measured with utmost care, all measurements are liable to have errors as discussed above dealing with large number of careful observations, we calculate the following errors.

Absolute Error :

Let a quantity be measured "n" times.

For the measured values to be $a_1, a_2 \ldots a_n$, the arithmetic mean is taken as the most accurate value of the quantity.

So, true value $= a_{mean} = \dfrac{a_1 + a_2 + \ldots + a_n}{n} = \dfrac{1}{n}\sum_{i=1}^{n} a_i$

Now we find the absolute errors by taking the magnitude of difference in individual readings and the true value as:

$$|\Delta a_1| = |a_1 - a_{mean}|$$

$$|\Delta a_2| = |a_2 - a_{mean}|$$

.....

.....

$$|\Delta a_n| = |a_n - a_{mean}|$$

Mean absolute error : The arithmetic mean of the individual absolute errors is called the mean absolute error in the measurement of the physical quantity 'a'.

$$\Delta a_{mean} = \frac{|\Delta a_1| + |\Delta a_2| + \ldots + |\Delta a_n|}{n}$$

$$= \frac{1}{n}\sum_{i=1}^{n} |\Delta a_i|$$

Thus, the value of the physical quantity "a" obtained in an experiment is written as

$$a = a_{mean} \pm \Delta a_{mean}$$

or $$(a_{mean} - \Delta a_{mean}) \le (a) \le (a_{mean} + \Delta a_{mean})$$

Relative Error :

Relative error or fractional error is given by,

$$\text{Relative error} = \frac{\Delta a_{mean}}{a_{mean}}$$

Percentage Error :

Percentage error is given by,

$$\delta a = \left(\frac{\Delta a_{mean}}{a_{mean}} \times 100 \right)\%$$

Q. **Find absolute error and percentage error in measurement of length of a rod. The readings are 2.48 m, 2.46 m, 2.49 m, 2.49 m and 2.46 m.**

Sol. Average length = Arithmetic mean of the measured value

$$x_{mean} = \frac{2.48 + 2.46 + 2.49 + 2.49 + 2.46}{5}$$

$$= \frac{12.38}{5} m$$

$$= 2.476 \text{ m}$$

$\therefore$ True value, x_{mean} = 2.48 m (rounding off is discussed in upcoming sections).

Absolute errors in various measurements.

$$|\Delta x_1| = |x_1 - x_{mean}| = 2.48 - 2.48 = 0.00\,m$$

$$|\Delta x_2| = |2.46 - 2.48| = 0.02\,m$$

$$|\Delta x_3| = |2.49 - 2.48| = 0.01 \text{ m}$$

$$|\Delta x_4| = |2.49 - 2.48| = 0.01 \text{ m}$$

$$|\Delta x_5| = |2.46 - 2.48| = +0.02 \text{ m}$$

$$\text{Mean absolute error} = \frac{|\Delta x_1| + |\Delta x_2| + |\Delta x_3| + \ldots + |\Delta x_5|}{5}$$

$$= \frac{(0.00 + 0.02 + 0.01 + 0.01 + 0.02)}{5}$$

$$= \frac{0.06}{5}$$

$$\Delta x_{mean} = 0.01 \text{ m}$$

Thus $x = 2.48 \pm 0.01$ m

Percentage error, $\delta x = \frac{\Delta x_{mean}}{x} \times 100$

$= \frac{0.01}{2.48} \times 100$

$= 0.40\%$

Q. **The time period of 100 oscillations of a simple pendulum is measured by a student 4 times.He gets the values 90s, 95 s, 92 s and 91 s. If the least count of the measuring clock is 1 s, then what should be the mean time reported by the student?**

(A) **92 ± 2 s**

(B) **92 ± 5.0 s**

(C) **92 ± 1.8 s**

(D) **92 ± 3s**

Sol. **(A)**

Least count of the clock = 1 s

Mean time, $\bar{t} = \frac{\sum_i t_i}{N}$

$\bar{t} = \frac{90 + 95 + 92 + 91}{4} = 92\,s$

Mean deviation $= \frac{\sum_i |\bar{t} - t_i|}{N} = \frac{2 + 1 + 3 + 0}{4} = 1.5\,s$

Since the least count of the instrument is 1 s, so reported mean time = (92 ± 2)s.

Combination of Errors

In many experiments, there are a number of quantities to be measured. These quantities are then to be put in some formula to get another quantity. For all the measured quantities in the experiment, there are some errors associated as well. Now, we are discussing how the errors are taken care of in case of combination of two or more quantities.

We can write an uncertain measurement in common notation as

$10\ m \leq L \leq 12\ m$

In more scientific notation, we can write it as,

$L = (11 \pm 1)\ m$

So, in general, for $a \leq L \leq b$,

$$L = \left(\underbrace{\frac{b+a}{2}}_{\text{Approximate value}(L_0)} \pm \underbrace{\frac{b-a}{2}}_{\text{Error}(\Delta L)} \right)$$

For $L = (L_0 \pm \Delta L)$

$$\text{Relative error} = \frac{\Delta L}{L_0}$$

$$\text{Percent relative error} = 100\left(\frac{\Delta L}{L_0}\right)\%$$

The above rule for finding errors is most basic way and it works for higher as well as lower errors. We will see some shortcuts for small errors in upcomg sections also, but those methods cannot be generalised for big values of errors.

Error in case of addition of quantities

Consider two quantities "a" and "b" are measured as given,

$a = 5\ kg \pm 2\ kg, \qquad b = 3\ kg \pm 1\ kg,$

Let a new quantity c is to be calculated as, c = a + b, then

$a_{max} = 5 + 2 = 7\ kg, \quad a_{min} = 5 - 2 = 3\ kg$

$b_{max} = 3 + 1 = 4\ kg, \quad b_{min} = 3 - 1 = 2\ kg$

$c_{max} = a_{max} + b_{max} = 7 + 4 = 11\ kg$

$c_{min} = a_{min} + b_{min} = 3 + 2 = 5\ kg$

$5\ kg \le c \le 11\ kg$

$$c = \left(\frac{11+5}{2}\right) \pm \left(\frac{11-5}{2}\right)$$

$$\boxed{c = (8 \pm 3)\ kg}$$

Clearly, for addition rule, if

$A = a \pm \Delta a$ and $B = b \pm \Delta b$

then, $c = A + B$

$$\mathbf{(c \pm \Delta c) = (a + b) \pm (\Delta a + \Delta b)}$$

Error in case of subtraction of quantities

Consider two quantities "a" and "b" are measured as given,

$a = 20 \pm 5$; $a_{max} = 25$; $a_{min} = 15$

$b = 10 \pm 5$; $b_{max} = 15$; $b_{min} = 5$

Let $c = a - b$, then,

$c_{max} = a_{max} - b_{min} = 25 - 5 = 20$

$c_{min} = a_{min} - b_{max} = 15 - 15 = 0$

$0 \le c \le 20$

so, $c = \left(\frac{20+0}{2}\right) \pm \left(\frac{20-0}{2}\right) = (10 \pm 10)$

Clearly, for subtraction rule, if

$A = a \pm \Delta a$, $B = b \pm \Delta b$

then, $c = A - B$,

$\mathbf{(c \pm \Delta c) = (a - b) \pm (\Delta a + \Delta b)}$

Here, the errors in individual quantities got added (not subtracted) in the final result.

Q. **If** $A = 10 \pm 1$, $B = 20 \pm 5$, $C = 30 \pm 10$

Then find $D = 3A + 2B - C$.

Sol.

$3A = 30 \pm 3$

$2B = 40 \pm 10$

$C = 30 \pm 10$

$D = 30 + 40 - 30 = 40$

$\Delta D = 3 + 10 + 10 = 23$

So, $(D \pm \Delta D) = (40 \pm 23)$

Error in case of multiplication of quantities

For two quantities "a" and "b" measured as,

$a = 4 \pm 2$; $a_{max} = 6$; $a_{min} = 2$

$b = 3 \pm 1$; $b_{max} = 4$; $b_{min} = 2$

For $c = (a \times b)$

$c_{max} = a_{max} \cdot b_{max} = 6 \times 4 = 24$

$c_{min} = a_{min} \cdot b_{min} = 2 \times 2 = 4$

$4 \le c \le 24$

$c \pm \Delta c = \left(\frac{24+4}{2}\right) \pm \left(\frac{24-4}{2}\right)$

$c = (14 \pm 10)$

So, in general if $A = a \pm \Delta a$

$B = b \pm \Delta b$

then, for $C = A \cdot B$

$$c = \left(ab + \underbrace{\Delta a \cdot \Delta b}_{\text{Neglect if very small}} \right) \pm (b \cdot \Delta a + a \cdot \Delta b)$$

Q. **Find the area of the given rectangle.**



l = (20 ± 5) cm
b = (10 ± 5) cm

Sol. $l_{max} = 25$ cm, $l_{min} = 15$ cm

$b_{max} = 15$ cm, $b_{min} = 5$ cm

$A_{max} = l_{max}\, b_{max} = 25 \times 15 = 375\ cm^2$

$A_{min} = l_{min}\, b_{min} = 15 \times 5 = 75\ cm^2$

$75\ cm^2 \le A \le 375\ cm^2$

$A_0 = \frac{75 + 375}{2} = 225\ cm^2$, $\Delta A = \frac{375 - 75}{2} = 150\ cm^2$

$A = A_0 \pm \Delta A = (225 \pm 150) cm^2$

Q. **Mass and speed of a particle are measured as follows –**

m = (4 ± 2) kg , v = (2 ± 1)m/s

The kinetic energy is best given by,

(A) (14 ± 13) J **(B) (8 ± 4) J**

(C) (8 ± 2) J **(D) (9 ± 2) J**

Sol. Kinetic energy, $K = \frac{1}{2}mv^2$

$m_{max} = 4 + 2 = 6$ kg ; $m_{min} = 4 - 2 = 2$ kg

$v_{max} = 2 + 1 = 3$ m/s ; $v_{min} = 2 - 1 = 1$ m/s

$K_{max} = \frac{1}{2} m_{max} v_{max}^2 = \frac{1}{2} \times 6 \times 3^2 = 27J$

$K_{min} = \frac{1}{2} m_{min} v_{min}^2 = \frac{1}{2} \times 2 \times 1^2 = 1J$

So, $1\ J \le K \le 27\ J$

$\therefore K = \left(\frac{27 + 1}{2}\right) \pm \left(\frac{27 - 1}{2}\right) = (14 \pm 13)\ J$

Error in case of division of quantities

Q. **Acceleration of a particle is calculated by measuring the net force acting on it and its mass. If**

$F = (20 \pm 5)$ N, $m = (10 \pm 5)$ kg,

the acceleration is given by –

(A) (2 ± 1) m/s² (B) (2 ± 0.7) m/s² (C) (3 ± 1) m/s² (D) (3 ± 2) m/s²

Sol. Acceleration, $a = \frac{F}{m}$

$F_{max} = 25$ N $m_{max} = 15$ kg

$F_{min} = 15$ N $m_{min} = 5$ kg

$$a_{max} = \frac{F_{max}}{m_{min}} = \frac{25}{5} = 5\ m/s^2$$

$$a_{min} = \frac{F_{min}}{m_{max}} = \frac{15}{15} = 1\ m/s^2$$

so, $$a = \left(\frac{5+1}{2}\right) \pm \left(\frac{5-1}{2}\right)$$

$$\boxed{a = (3 \pm 2)\, m/s^2}$$

The differentiation method for small errors : Examples

Q. **Let a number $c = a^3 b^2$**

If % age error in a is 1 % and %age error in b is 2 %, then find %age error in c.

Sol. $c = a^3 b^2$

Taking ln on both sides,

$\ln c = \ln(a^3 b^2)$

$\ln c = \ln a^3 + \ln b^2$

$\ln c = 3 \ln a + 2 \ln b$

Differentiate both sides wrt x,

$$\frac{d}{dx}(\ln c) = \frac{d}{dx}(3 \ln a) + \frac{d}{dx}(2 \ln b)$$

$$\frac{1}{c} \cdot \frac{dc}{dx} = 3\left(\frac{1}{a}\right)\frac{da}{dx} + 2\left(\frac{1}{b}\right)\frac{db}{dx}$$

$$\Rightarrow \frac{dc}{c} = 3\left(\frac{da}{a}\right) + 2\left(\frac{db}{b}\right)$$

When errors are small, we can say $\frac{dc}{c} = \frac{\Delta c}{c}, \frac{da}{a} = \frac{\Delta a}{a}, \frac{db}{b} = \frac{\Delta b}{b}$.

$$\therefore \quad \left(\frac{dc}{c} \times 100\%\right) = 3\left(\frac{da}{a} \times 100\%\right) + 2\left(\frac{db}{b} \times 100\%\right)$$

$$= 3\,(1\,\%) + 2(2\,\%)$$

%age error in $c, \left(\frac{dc}{c} \times 100\,\%\right) = 7\,\%$

In general, if $z = \frac{a^m, b^n}{c^q}$, then the maximum relative error in z is given by

$$\frac{\Delta z}{z} = \frac{m\Delta a}{a} + \frac{n\Delta b}{b} + q\frac{\Delta c}{c}$$

Q. **Time period of a simple pendulum is given by $T = 2\pi\sqrt{\frac{l}{g}}$.**

If %age error in measurement of l is 2% and %age error in measurement of g is 3%, then find %age error in T.

Sol. $T = 2\pi\sqrt{\frac{l}{g}}$

$$\ln(T) = \ln\left(2\pi\sqrt{\frac{L}{g}}\right) = \ln 2\pi + \ln\sqrt{l} - \ln\sqrt{g}$$

$$\ln(T) = \ln(2\pi) + \frac{1}{2}\ln(l) - \frac{1}{2}\ln(g)$$

Differentiating both sides,

$$\frac{1}{T}dT = 0 + \frac{1}{2}\frac{dl}{l} - \frac{1}{2}\left(\frac{dg}{g}\right)$$

Convert this –ve sign to +ve as errors can never get subtracted.
If errors are small, then,

$$\%\text{age error in } T = \left(\frac{dT}{T} \times 100\%\right) = \frac{1}{2}(2\%) + \frac{1}{2}(3\%)$$

$$= \left(1 + \frac{3}{2}\right)\% = 2.5\,\%$$

Q. **Find %age error in $D = \frac{a^2 b^{\frac{1}{2}}}{c^3}$**

%age errors in a, b and c are 1%, 3% and $\frac{1}{2}$% respectively.

Sol. %age error in $D = 2(1\%) + \frac{1}{2}(3\%) + 3\left(\frac{1}{2}\%\right)$

$= 2 + 1.5 + 1.5 = 5\%$

Q. **$y = 20\,e^{\frac{x}{100}}$**

There is 1 % error in x. Find % error in y when x = 200.

Sol.

$$y = 20\,e^{\frac{x}{100}}$$

$$\frac{dy}{dx} = 20\left(\frac{1}{100}\right)e^{\frac{x}{100}}$$

$$dy = \frac{1}{5}e^{\frac{x}{100}} \cdot dx$$

Divide by y on both sides,

$$\frac{dy}{y} = \frac{1}{5}e^{\frac{x}{100}} \cdot \frac{dx}{y}$$

$$\% \text{ error in y, } \left(\frac{dy}{y}\right) \times 100\% = \left[\frac{1}{5}e^{\frac{x}{100}} \cdot \frac{dx}{20\,e^{x/100}}\right] \times 100\%$$

$$= \left(\frac{1}{100} \cdot dx\right) \times 100\%$$

$$= \frac{1}{100} \cdot \left(\frac{dx}{x}\right) \times 100\% \times x$$

$$= \frac{1}{100}\left[1\%\right] \times 200 \qquad \text{...(Given that x = 200)}$$

$$= 2\ \%$$

Q. **$y = a^2b + c^2$**
% errors in a, b, c are 1 %, 2 % and 1 % respectively.
Find % error in y when a = b = c = 10 units.

Sol. Here c^2 is added to a^2b. So, we have to start from beginning and taking ln on both sides will not help here.

$$y = a^2b + c^2$$

Differentiating both sides,

$$dy = d(a^2 b) + d(c^2)$$

$$dy = (2a\,b \cdot da + a^2 \cdot db) + 2c \cdot dc$$

$$(dy) \times 100\% = \left[2a^2 b\left(\frac{da}{a}\right) + a^2 b\left(\frac{db}{b}\right) + 2c^2\left(\frac{dc}{c}\right)\right] \times 100\,\%.$$

$$100\,dy = 2a^2b\,[1\,\%] + a^2b\,[2\,\%] + 2c^2[1\,\%]$$

$$= 2(10)^2\,(10)\,(1\,\%) + 10^2\,(10)\,(2\,\%) + 2\,(10)^2\,(1\,\%)$$

$$100\,dy = 2000 + 2000 + 200$$

$$100\left(\frac{dy}{y}\right) = \frac{4200}{y}$$

$$= \frac{4200}{a^2 b + c^2}$$

$$= \frac{4200}{10^2\,(10) + 10^2} = \frac{4200}{1100}$$

$$= \frac{42}{11}$$

$$\boxed{\left(\frac{dy}{y}\right) 100\,\% = \frac{42}{11}\%}$$

An interesting concept :

Find equivalent resistance for two resistances in parallel. Resistances are -

$R_1 = (30 \pm 0.3)\ \Omega$

$R_2 = (60 \pm 0.3)\ \Omega$

If we find the percentage error by the addition formula,

For resistors in parallel,

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$R_1 + R_2 = (30 + 60) \pm (0.3 + 0.3)$$

$$= (90 \pm 0.6)\Omega$$

$$\% \text{ error in } (R_1 + R_2) = \frac{0.6}{40} \times 100\% = \frac{2}{3}\%$$

$$\% \text{ error in } R_1 = \frac{0.3}{30} \times 100\% = 1\%$$

$$\% \text{ error in } R_2 = \frac{0.3}{60} \times 100\% = \frac{1}{2}\%$$

$$\% \text{ error in } R_{eq} = (\% \text{ error in } R_1) + (\% \text{ error in } R_2) + (\% \text{ error in } R_1 + R_2)$$

$$= 1\% + \frac{1}{2}\% + \frac{2}{3}\% = \frac{13}{6}\%$$

Now, if we find it by the differentiation method,
We know that, for parallel connection,

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

Differentiating both sides,

$$d\left(\frac{1}{R_{eq}}\right) = d\left(\frac{1}{R_1}\right) + d\left(\frac{1}{R_2}\right)$$

$$-\frac{1}{R_{eq}^2} dR_{eq} = -\frac{1}{R_1^2} dR_1 - \frac{1}{R_2^2} dR_2$$

$$\Rightarrow \quad \frac{1}{R_{eq}}\left(\frac{dR_{eq}}{R_{eq}}\right) = \frac{1}{R_1}\left(\frac{dR_1}{R_1}\right) + \frac{1}{R_2}\left(\frac{dR_2}{R_2}\right)$$

$$\frac{1}{R_{eq}}\left[\% \text{ error in } R_{eq}\right] = \frac{1}{R_1}(1\%) + \frac{1}{R_2}\left(\frac{1}{2}\%\right)$$

$$\% \text{ error in } R_{eq} = \left[\left(\frac{1}{30}\right)\% + \left(\frac{1}{120}\right)\%\right] R_{eq}$$

$$R_{eq} = \frac{30 \times 60}{(30 + 60)} = 20\,\Omega$$

$$\% \text{ error in } R_{eq} = \left(\frac{2}{3} + \frac{1}{6}\right)\% = \left(\frac{4+1}{6}\right)\% = \frac{5}{6}\%$$

Clearly we get different results by using different methods. The catch here is to use the formula in lowest simple form to reduce the error.

The addition formula which we used was not the lowest simple form of the formula for equivalent resistance. Hence the second method used is the correct one.

$\frac{1}{R_1} + \frac{1}{R_2}$ is lowest simple form.

$\frac{R_1 R_2}{R_1 + R_2}$ magnifies the error, Any fraction should be reduced to its lowest form before applying rules of error calculations.

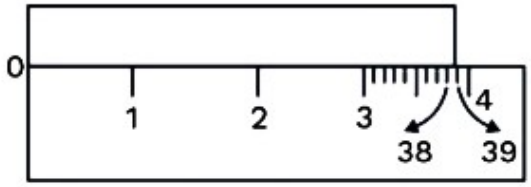
For example, if $z = y\frac{(x^2 - 1)}{(x + 1)}$

Then, we have to simply this first before doing error analysis.

$$z = y\frac{(x-1)(x+1)}{(x+1)} = y(x-1). \qquad \rightarrow \text{Lowest form.}$$

Relation between Error and least count of an instrument

If we take a measurement using a scale and we get the reading as shown in the below figure,



The measured length is lying somewhere between 38 and 39 and we cannot exactly tell the value, then we report the measurement as,

$l = (38.5 \pm 0.5)$ mm

where $\boxed{\Delta l = \frac{\text{L.C.}}{2}}$, L.C. is the least count of the instrument.

In some books, $\boxed{\Delta l = L \cdot C}$

Both these considerations for Δl are technically correct.

In exams, we generally take Δl = L.C.

Q. **Length and breadth of a rectangle as measured by a mm scale are 100 mm and 50 mm respectively. Find the percent error in its area.**

Sol. As the measurement is done with mm scale, so,

$L = 100 \text{ mm} \pm 1 \text{ mm}$

$b = 50 \text{ mm} \pm 1 \text{ mm}$

$$\% \text{ error in } L = \frac{1}{100} \times 100 = 1\%$$

$$\% \text{ error in } b = \frac{1}{50} \times 100 = 2\%$$

Area, $A = Lb$

% error in area = (1 + 2) % = 3 %

Significant Figures and rounding off

All measurements are approximate. So, reporting of a measurement indicates the precision of the instrument used for the measurement. For a measurement of 11.27 cm using a vernier calliper, the measurement is done upto a resolution of 0.01 cm. We can also say that the uncertainty in this measurement is ± 0.01 cm. It shows that the actual value will lie between 11.26 cm to 11.28 cm. So, in the reported value 11.27 cm, the digits 1,1 and 2 are certain digits while the digit 7 is uncertain. All the certain digits and one uncertain digit in a reported value of a measurement are collectively known as significant digits. So, 11.27 cm has 4 significant digits.

Significant digits in a measurement

S. No.	Number	Number of significant digits	Rule/comments
1.	2165.17	6	All non-zero digits must be counted as significant digits.
2.	145.00	5	Zeroes on the right are significant.
3.	00.00045	2	Zeroes on the left are not significant.
4.	23.00003	7	Zeroes between non-zero numbers are significant.
5.	11.04×10^6	4	Only the significant digits in mantissa are counted.
6.	4.2 E – 6	2	4.2×10^{-6}
7.	e, π, 2	Infinite	Exact mathematical constants, numbers have infinite significant digits.
8.	1 inch = 2.54 cm	Infinite	Defined conversion factors (like 1 kg = 1000 gm) have infinite significant digits.
9.	1 lb = 453.5923 gm (measured)	7	Experimentally measured conversion factors do not have infinite significant figures.
10.	1200 meters	2 or 3 or 4	The result of measurement must be written in scientific notation to avoid ambiguity.
11.	A value and its uncertainly should have the same units, exponent, and number of significant figure. e.g. c = $(2.99792 \pm 0.00030) \times 10^8$ m/s		

Q. **Based on the above mentioned rules in the table, find the number of significant digits in given measurements.**

a. 5.0700	**b. 0.00564**	**c. 3.07×10^{-4}**	**d. 21400**	**e. 0.005003**
f. 4.700×10^{9}	**g. 150**	**h. 65,00,000**	**i. 0.0204**	**j. 0.00600**

Sol.

a. In 5.0700, there are 5 significant figures.
b. In 0.00564, there are 3 significant figures.
c. In 3.07×10^{-4} , there are 3 significant figures.
d. 21400 is an ambiguous case. There can be 3 or 4 or 5 significant figures.
e. In 0.005003, there are 4 significant figures.
f. In 4.700×10^{9} , there are 4 significant figures.
g. 150 is again ambiguous case. There can be 2 or 3 significant figures.
h. In 65,00,000, there are minimum 2 significant figures. It is also an ambiguous case.
i. In 0.0204, there are 3 significant figures.
j. In 0.00600, there are 3 significant figures.

Rounding off numbers

Case A:	
In rounding off numbers, the last figure kept should be unchanged if the first figure dropped is less than 5.	e.g. If only one decimal is to be kept, then 3.422 becomes 3.4.
Case B:	
In rounding off numbers, the last figure kept should be increased by 1 if the first figure dropped is greater than 5.	e.g. If only two decimals are to be kept, then 4.3872 becomes 4.39. Similarly, 2.998 because 3.00.
Case C:	
In rounding off numbers, if the first figure dropped is 5, and all the figures following the five are zero or if there are no significant figures after the 5, then the last figure kept should be unchanged if that last figure is even.	e.g. If only one decimal is to be kept, then 6.6500 becomes 6.6. If only 2 decimals are to be kept, then 7.485 becomes 7.48.

Case D:

In rounding off numbers, if the first figure dropped is 5, and all the figures following the 5 are zero or if there are no figures after the 5, then the last figure kept should be increased by 1 if that last figure is odd.

e.g. If only two decimals are to be kept, then 6.755000 becomes 6.76. If only two decimals are to be kept, then 8.995 becomes 9.00.

Case E:

In rounding off numbers, if the first figure dropped is 5, and there are any figures following the 5 that are not zero, then the last figure kept should be increased by 1.

e.g. If only one decimal is to be kept, then 6.6501 becomes 6.7.
If only two decimals are to be kept, then 7.4852007 becomes 7.49.

Q. **Round off 4.17493 to 3 significant digits.**

Sol. We have to round off 4.17493 to 3 significant digits. So, we have to remove 493. As 493 < 500, so using case A, answer will be 4.17 rounded off to 3 significant figures.

Q. **Round off each of the following numbers to 2 decimal places.**

1. 17.045	**2. 257.135**	**3. 948.3349**
4. 19.04501	**5. 482.005**	**6. 482.00523**
7. 0.015	**8. 2.495**	

Sol.

1. In 17.045, the number to be dropped is 5 to round off upto 2 decimal places. As 4 is even, so we leave it unchanged and the answer is 23.04.
2. In 257.135, the number to be dropped is 5 to round off upto 2 decimal places. As 3 is odd, so we we increase it by 1 and the answer is 257.14.
3. In 948.3349, 49 is to be rounded off which is less than 50, so we leave 33 unchanged. So, the answer is 948.33 (not 948.34).
4. In 19.04501, 501 is more than 500, so we increase 4 by 1. The answer is 19.05 (not 19.04).
5. In 482.005, to round off upto 2 decimal places, there are no zeroes or digits after 5, so we increase the last digit to be kept by 1. The answer is 482.01.
6. By dropping 523, we increase the digit preceding 5 by 1. Answer is 482.01.
7. By dropping 5, we increase preceding digit by 1. The answer is 0.02.
8. By dropping 5, we increase the preceding digit by 1. Answer is 2.50 (not 2.5).

Calculations with significant digits : Multiplication/Division

The rough rule for multiplication and division is –

1. Count the number of significant digits of the numbers involved.
2. If the number with minimum significant digits has 'η' significant digits, then the answer should be rounded off to 'η' significant digits.

Note: Regular significant figure rules are guidelines, and they don't always predict the correct number of significant figures.

We use the significant figures rules as a shortcut so that we don't have to do a detailed error analysis on everything we calculate.

Q. **Length and breadth of a rectangular frame is given as**
L = 27.93 cm and B = 20.36 cm.
The area of the sheet is best given by,
(A) 568.6547 cm² (B) 568.6 cm² (C) 568.7 cm² (D) 568.65 cm²

Sol. Area, A = LB = 27.93 × 20.36
= 568.6548 cm²
Answer should have 4 significant figures only.
568.6548 → 548 > 500
so, 568.7 is the answer.

Calculations with significant digits : Addition/Subtraction

The rough rule for addition and subtraction is-

1. Perform addition subtraction as usual.
2. If the number with minimum number of digits after decimal has 'η' digits after decimal, then the answer should be rounded off to 'η' digits after decimal.

Note: The answer can have more significant digits than the input with least number of significant figures.

Q. **Add 1.2 and 4.41 upto appropriate significant figures.**

Sol. 1.2 + 4.41 = 5.61
Limit answer to 1 decimal place only, so, 5.6 is answer.

Q. **Add 77.2 and 10.46 upto appropriate significant figures.**

Sol.
$$\begin{array}{r} 77.2 \\ +\ 10.46 \\ \hline 87.66 \end{array} \rightarrow 87.7$$

An interesting case

Multiply a number by an integer such as 1.234 × 9.

By multiplication rule : 1.234 × 9.000..... = 11.106 = 11.11

By addition rule:

Add 1.234 to itself 9 times such as,

1.234 + 1.234 + ---------- + 1.234 = 11.106

(One significant digit increases)

Addition rule is considered more fundamental.

so, 11.106 is more correct answer.

Mixed Operations

(1) 2.8 × 4.532 + 12.690

= 12.6896 + 12.690 .....(Do not round off intermediate values, but 12.6896 must have 2 significant figures only)

= 25.3796 .....(This is written here, but it has 2 sig. figs only)

= 25 .....(Round off to zero decimal places)

(2) (15.803 – 4.76)/9.3

= (11.043)/9.3

= 1.18742

= 1.2

(3) 3 + 15/12 = 3 + 1.25 = 4.25 = 4

(4) (1.07 – 0.8826)/0.762 = 0.1874 / 0.762 = 0.2459 = 0.25

Although, the result of the subtraction is written as 0.1874, but it should have the significant digits up to two decimal places only.

Calculations with significant digits: Logs and exponents

When we take the log of a number with N significant figures, the result should have N decimal places.

log (0.0056) = - 2.2518 = -2.25

log (0.00724) = - 2.14026 = -2.140

When raising 10 to a power:

If the power of 10 has N decimal places, the result should have N significant figures. The rule for raising e to a power is similar. For simplicity, we will use this rule for both situations.

$10^{1.2} = 15.8 = 20 = 0.2 \times 10^2$

$10^{1.20} = 15.8 = 16 = 1.6 \times 10^1$

$10^{1.2000} = 15.80$

$10^{1.3} = 19.9 = 20 = 0.2 \times 10^2$

$10^{1.4} = 25.1 = 30 = 0.3 \times 10^2$

$10^{2.890} = 776.25 = 776$

$10^{2.89} = 776.25 = 780 = 7.8 \times 10^2$.

Dimensional Formulae

S.NO.	PHYSICAL QUANTITY	RELATION WITH OTHER QUANTITIES	DIMENSIONAL FORMULA	SI UNIT
1.	Area	Length × breadth	$L \times L = L^2 = [M^0L^2T^0]$	m^2
2.	Volume	Length × breadth × height	$L \times L = L = [M^0L^3T^0]$	m^3
3.	Density	Mass/Volume	$\frac{M}{L^3} = [ML^{-3}T^0]$	$kg\,m^{-3}$
4.	Speed or velocity	$\frac{\text{Distance}}{\text{Time}}$	$\frac{L}{T} = [M^0LT^{-1}]$	ms^{-1}
5.	Acceleration	$\frac{\text{Change in velocity}}{\text{Time}}$	$LT^{-2} = [M^0LT^{-2}]$	ms^{-2}
6.	Momentum	Mass × velocity	$[MLT^{-1}]$	$kg\,ms^{-1}$
7.	Force	Mass × acceleration	$[M\,L\,T^{-2}]$	N
8.	Work	Force × distance	$MLT^{-2} \times L = [ML^2T^2]$	J
9.	Energy	Amount of work	$[ML^2T^{-2}]$	J
10.	Power	$\frac{\text{Work}}{\text{Time}}$	$\frac{ML^2T^{-2}}{T} = [ML^2T^{-3}]$	W
11.	Pressure	$\frac{\text{Force}}{\text{Area}}$	$\frac{ML^1T^{-2}}{L^2} = [ML^{-1}T^{-2}]$	Pa or Nm^{-2}
12.	Moment of force/ Torque	Force × Perpendicular distance	$MLT^{-2} \times L = [ML^2T^{-2}]$	Nm
13.	Gravitational constant	$\frac{\text{Force} \times (\text{distance})^2}{(\text{Mass})^2}$	$[M^{-1}L^3T^{-2}]$	$N\,m^2\,kg^{-2}$
14.	Impulse of a force	Force × time	$[MLT^{-1}]$	N s

S.NO.	PHYSICAL QUANTITY	RELATION WITH OTHER QUANTITIES	DIMENSIONAL FORMULA	SI UNIT
15.	Stress	$\frac{\text{Force}}{\text{Area}}$	$\frac{MLT^{-2}}{L^2} = [ML^{-1}T^{-2}]$	$N\ m^{-2}$
16.	Strain	$\frac{\text{Change in dimension}}{\text{Original dimension}}$	$[M^0L^0T^0]$	No unit
17.	Coefficient of elasticity	$\frac{\text{Stress}}{\text{Strain}}$	$\frac{ML^{-1}T^{-2}}{1} = [ML^{-1}T^{-2}]$	$N\ m^{-2}$
18.	Surface tension	$\frac{\text{Force}}{\text{Length}}$	$\frac{MLT^{-2}}{L} = [ML^0T^{-2}]$	$N\ m^{-1}$
19.	Surface energy	$\frac{\text{Energy}}{\text{Area}}$	$\frac{ML^2T^{-2}}{L^2} = [ML^0T^{-2}]$	$J\ m^{-2}$
20.	Coefficient of viscosity	$\frac{\text{Force} \times \text{distance}}{\text{Area} \times \text{velocity}}$	$\frac{MLT^{-2} \times L}{L^2 \times LT^{-1}} = [ML^{-1}T^{-1}]$	$Ns\ m^{-2}$
21.	Angle	$\frac{\text{Arc}}{\text{Radius}}$	$\frac{L}{L} = 1 = [M^0L^0T^0]$	radians
22.	Angular velocity	$\frac{\text{Angle}}{\text{Time}}$	$\frac{1}{T} = T^{-1} = [M^0L^0T^{-1}]$	$rad\ s^{-1}$
23.	Angular acceleration	$\frac{\text{Angular velocity}}{\text{Time}}$	$\frac{T^{-1}}{T} = T^{-2} = [M^0L^0T^{-2}]$	$rad\ s^{-2}$
24.	Moment of inertia	Mass × (distance)^2	$ML^2 = [ML^2T^0]$	$Kg\ m^2$
26.	Angular momentum	Mass × velocity × radius	$[ML^2T^{-1}]$	$kg\ m^2s^{-1}$
27.	Frequency	$\frac{1}{\text{Time period}}$	$\frac{1}{T} = T^{-1} = [M^0L^0T^{-1}]$	Hz or s^{-1}
28.	Plank's constant	$\frac{E}{\nu} = \frac{\text{Energy}}{\text{Frequency}}$	$\frac{ML^2T^{-2}}{T^{-1}} = [ML^2T^{-1}]$	Js

S.NO.	PHYSICAL QUANTITY	RELATION WITH OTHER QUANTITIES	DIMENSIONAL FORMULA	SI UNIT
29.	Velocity gradient	$\frac{\text{Velocity}}{\text{Distance}}$	$\frac{LT^{-1}}{L} = T^{-1} = [M^0L^0T^{-1}]$	s^{-1}
30.	Pressure gradient	$\frac{\text{Pressure}}{\text{Distance}}$	$\frac{ML^{-1}T^{-2}}{L} = [ML^{-2}T^{-2}]$	Pa m^{-1}
31.	Boltzmann's constant	$\frac{\text{Energy}}{\text{Temperature}}$	$\frac{[ML^2T^{-2}]}{[K]} = [ML^2T^{-2}K^{-1}]$	$J\ K^{-1}$
32.	Thermal conductivity	$\frac{\text{Heat} \times \text{distance}}{\text{Area} \times \text{temp.} \times \text{time}}$	$\frac{[ML^2T^{-2}L]}{[L^2.K.T]} = [MLT^{-3}K^{-1}]$	$Js^{-1}\ m^{-1}\ K^{-1}$
33.	Electric charge	Coulomb=Time × Current	$[T.A] = [M^0L^0TA]$	C
34.	Electrical potential	$\frac{\text{Work}}{\text{Charge}}$	$\frac{[ML^2T^{-2}]}{[TA]} = [ML^2T^{-3}A^{-1}]$	V (volt)
35.	Resistance	$\frac{\text{Potential difference}}{\text{Current}}$	$\frac{[ML^2T^{-3}A^{-1}]}{[A]} = [ML^2T^{-3}A^{-2}]$	W (ohm)
36.	Capacitance	$\frac{\text{Charge}}{\text{Potential difference}}$	$\frac{[TA]}{[ML^2T^{-3}A^{-1}]} = [M^{-1}L^{-2}T^4A^2]$	F (farad)
37.	Inductance	$\frac{\text{EMF}}{\text{current / time}}$	$\frac{[ML^2T^{-3}A^{-1}]}{[AT^{-1}]} = [ML^2T^{-2}A^{-2}]$	H (henry)